# Closing the 89-element endpoint

Leading-core overlap and exact covering certificates

Sum-of-Subset research project

18 October 2026

**Abstract**

We strengthen the necessary covering model for a hypothetical 89-element set by counting the overlap between a leading $2D$ component and an own $D$ translate. Four new inequalities exclude the previously published exact relaxation witness, each by exactly 72. We extract this mechanism from the archived size-89 proof, reprove its actual-set hypotheses and loss allocation, and generate new certificates. A whole-side deficit also retains target cardinality without double counting overlapping subgroups. Final verification statistics and the endpoint conclusion are in Section 5. This continues the independently audited chain ending at 89; the stronger archived claim 84 remains a separate audit task.

## 1 Endpoint reduction and structural domain

For a finite real set $A$, $\mathrm{FS}(A)$ includes the empty sum and $4A$ allows repeated summands. Let $N$ be the largest possible size with $\mathrm{FS}(A) \subseteq 4A$. The previous independently checked chain proves $71 \le N \le 89$ [1, 3]. It suffices to exclude size 89. Such a set contains zero: otherwise adjoining zero preserves feasibility and gives a forbidden 90-element set. Integralization preserves the zero coordinate, distinctness, order and the chosen subset-sum identities. No arbitrary deletion is asserted to preserve feasibility.

The preceding progress report [2] proves the structural reduction for this endpoint, independently of its incomplete covering search. Its nested sets satisfy

$$|D| = 81, \quad |E| = 73, \quad E = P \cup \{0\} \cup (-Q), \quad |P| = a, \quad |Q| = b, \quad a + b = 72.$$

On the common primitive core lattice, $D$ is integral. With

$$I = (a - 36)^2, \quad \Lambda = 1333 + I, \quad \Omega = 546 - I, \quad C = -883 - I,$$

the inherited exact budgets are

$$\rho_P + \rho_Q + 2\vartheta + c + \mathcal{T} = 546 - I, \tag{1}$$

$$\mathcal{T} \ge \kappa + 152 - 3z, \qquad \mathcal{T} \ge \max\{51, z, 152 - 2z\}, \tag{2}$$

$$c = 2|2D| + 18 \cdot 81 + 85 - |\,\mathrm{FS}(D)|. \tag{3}$$

Here $\kappa = |\,\mathrm{FS}(P) - \mathrm{FS}(Q)| - |\,\mathrm{FS}(P)| - |\,\mathrm{FS}(Q)| + 1$, $z = h_P + h_Q - \kappa \ge 0$, $h_P, h_Q$ are side-support hull holes, and all slack terms are nonnegative.

The hypotheses of the inherited structural proof remain intact. Intrinsic primitive normalization repairs sixteen insufficient inverse directions. Early prefix retention proves the remaining inner lattice comparison before (2) is used. The small-side boundary uses the universal negative prefix floor, not the size-17 non-AP theorem. Progression and half-step cases are separately handled; surviving progression cases have $z = 0$. Two retained prefixes give the positive-hole mass and span rows. In the sharpened R budget, defect and peeling estimates retain the same hole parameter. The low-hole domain is

| Band | Hole range | Cap for $c$ | Empty units $j$ |
|---|---|---|---|
| Z | $0$ | $394 - I$ | $0, \dots, 4$ |
| P | $1, 2$ | $336 - I$ | $0, \dots, 4$ |
| Q | $3, 4$ | $328 - I$ | $0, \dots, 4$ |
| R | $5, \dots, 14$ | $313 - I$ | $0, \dots, 3$ |

High-hole states retain the coupled $z$ variable and all three caps in (2). No separately maximized caps and holes are combined.

Write $M_p = \max D$, $M_q = -\min D$, $R_p = \max P$, $R_q = \max Q$. The four outer positive coordinates satisfy $p_1 > p_2 > p_3 > p_4 > M_p$, and similarly on the negative side. The inherited integer-row-lattice test is retained, including the thirty four-empty patterns without complete outer integrality. The new rows are inserted only when that test proves complete outer integrality. No nonunit pivot is divided away. The complete root domain, including these exceptional patterns and all permitted actual paired contributors, is regenerated by the verifier.

## 2 The missing raw overlap

On the positive covering side put

$$c_p = p_1 - p_2 - p_3 - p_4, \qquad U_0 = c_p + 2D, \qquad U_i = c_p + p_i + D.$$

The first is the leading component, the second a retained own component; both are considered before intersection with the actual target $F_p$. Put $K = |2D|$ and $d = 81$.

**Lemma 2.1** (Leading-core intersection)**.** *If $p_i - M_p$ is integral, then*

$$|U_0 \cap U_i| \geq H_i := 2|E| - R_p - R_q - p_i + M_p - 1.$$

*Proof.* The leading component contains $c_p + M_p + E$, since $M_p \in D$ and $E \subseteq D$. The own component contains $c_p + p_i + E$. After translating by $c_p + M_p$, these are two integral sets of size $|E|$ inside an integer interval with $R_p + R_q + p_i - M_p + 1$ points. Inclusion-exclusion proves the bound, including when $H_i < 0$. Neither leading contribution nor target fullness is required. □

**Lemma 2.2** (One outside loss at a time)**.** *Let $L_0 = |U_0 \setminus F_p|$ and $L_i = |U_i \setminus F_p|$. Then*

$$K + d - |(U_0 \cup U_i) \cap F_p| \geq H_i + \max\{L_0, L_i\}.$$

*Proof.* The left side equals

$$|U_0 \cap U_i| + |(U_0 \cup U_i) \setminus F_p|.$$

The first term is at least $H_i$ and the second dominates either individual outside loss. Adding both outside losses is invalid: for $U_0 = U_i = \{0, 1\}$ and $F_p = \{1\}$ the deficit is 3, whereas $|U_0 \cap U_i| + L_0 + L_i = 4$. □

Reflection replaces $D$ by $-D$ and interchanges $p, q$. This is raw overlap; no extra outside loss is added to a clipped bound.

## 3 Rows and simultaneous loss allocation

The old variable $G_p$, named `full_p`, has an actual realization. Group the leading component with the retained own slots $p0, p1, p2, p3$ and cross slots $p6, p7$; let $J_p$ denote these ordinary labels. Set

$$G_p = K + d|J_p| - \left| \left( U_0 \cup \bigcup_{v \in J_p} U_v \right) \cap F_p \right|.$$

For an ordinary label $v$, $\lambda_v = l_v + u_v$ counts lower and upper hull-tail losses, hence does not exceed the actual loss outside $F_p$, also when $F_p$ has holes. The model's $L_p$ (`lead_p`) does not exceed $L_0$. Bound the larger union by one pair union and the remaining individual members. For every retained own slot $v = p(i-1)$ this gives

$$G_p \geq H_i + L_p + \sum_{w \in J_p \setminus \{v\}} \lambda_w, \tag{4}$$

$$G_p \geq H_i + \lambda_v + \sum_{w \in J_p \setminus \{v\}} \lambda_w. \tag{5}$$

These are alternative lower bounds for one deficit, not independent savings to be summed. The $q$ group has disjoint labelled components, so both sign groups can constrain the global budget simultaneously.

The implementation copies the old row $-G_p + L_p + \sum_{J_p} \lambda_v \leq 0$. For (4) it deletes the selected own loss; for (5) it deletes $L_p$. It then adds $-R_p - R_q - p_i + M_p$ and uses right side $1 - 2|E| = -145$. Slot $p(i-1)$ is mapped to outer coordinate $p_i$. The negative side is identical. The rows have no $\delta$ coefficient and recharge no fixed-label losses. Their new coefficients and right sides are independent of $a, z, S_p, S_q$; the previous positive-hole mass transport and high-state grouping preserve them unchanged.

### 3.1 Whole-side capacity in the two special modes

In the Z4 modes with both paired units and both largest own slots absent, each side has seven retained, unpaired ordinary members $V_s$. Define

$$W_s = K + 7d - \left| \left( U_{0,s} \cup \bigcup_{v \in V_s} U_v \right) \cap F_s \right|, \qquad s = p, q.$$

Because $F_s \subseteq [1, S_s]_{\mathbb{Z}}$ and union cardinality is subadditive,

$$W_s \geq K + 7d - S_s, \tag{6}$$

$$W_s \geq L_s + \sum_{v \in V_s} \lambda_v, \tag{7}$$

$$W_s \geq G_s + \sum_{v \in V_s \setminus J_s} \lambda_v. \tag{8}$$

One actual $W_s$ satisfies all three rows. The first uses target cardinality; the others partition the labelled components into disjoint families. The last row charges the $G_s$ subgroup only once.

Put $\delta = c - 81j$. The inherited 110-score construction gives eight mandatory high-label losses and a disjoint extra fixed loss $F_{\text{extra}}$. The full covering budget is

$$W_p + W_q + 8 + F_{\text{extra}} \leq \delta.$$

It uses each whole-side group once; $G_p, G_q$, their subgroups and the same ordinary losses are not separately added. The old budgets remain valid alternatives under this simultaneous actual realization. The fixed-score catalogue, all integer threshold alternatives, the 26 disjoint nonpositive deletion groups, and exclusions of separately charged high labels are unchanged.

## 4 Removal of the former obstruction

The exact rational point in `data/upper88_relaxation_witness.json` remains a valid witness for the preceding relaxation, not for an actual feasible set. It has

$$R_p = R_q = 36, \quad p_4 = 122, \quad M_p = 121, \quad q_4 = 124, \quad M_q = 123, \quad G_p = G_q = 1.$$

For slot $p3$,

$$H_4 = 146 - 36 - 36 - 122 + 121 - 1 = 72.$$

The other losses in its full group sum to 1, and the selected own and leading losses are zero. Each of (4) and (5) therefore requires $G_p \geq 73$, contradicting $G_p = 1$ by 72. The negative side gives two further violations of 72. This exact calculation explains the removed obstruction without appealing to a numerical infeasibility status.

All parameters $28 \leq a \leq 44$ of this formerly obstructed A-mode close. Only $a = 34, 35, 36, 37, 38$ need binary trees in the discovered certificate; the others admit direct integer Farkas certificates. This local result is separate from the complete root-universe check below.

## 5 Exact verification and conclusion

The verifier reconstructs the inherited root universe, augments only eligible nested models, and checks both children of each proved binary disjunction. Coarse rows and the disjunction catalogue are unchanged. For nonnegative model variables and $Ax \leq b$, an accepted integer certificate satisfies

$$y \geq 0, \qquad y^T A \geq 0, \qquad y^T b < 0.$$

Parameter intervals use exact quadratic polynomials, checked at endpoints and the integers adjacent to any relevant interior vertex. No polynomial is inferred from floating-point samples. Exhaustive mode regenerates each integer matrix and compares every row with its symbolic version.

| Verification quantity | Exact value |
|---|---|
| Structural root groups | 61,884 |
| Integer root systems | 641,524 |
| Coarse / nested groups | 60,766 / 1,118 |
| Integer certificate vectors | 2,773 |
| Quadratic interval checks | 61,980 |
| Checked extremum points | 164,353 |
| Integer roots using binary trees | 55 |
| Binary splits / tree leaves | 95 / 150 |
| Maximum tree depth | 10 |
| Final certified systems | 641,619 |
| Minimum normalized strict gap | 11/1069 |

Every reconstructed root is covered, with no omitted integer parameter or disjunction child. The split types are 58 core-loss, 20 ordinary-position, 6 outer-pair and 11 core-pair-order splits. The complete symbolic check and the full integer-matrix replay both passed. Standard and `-O` outputs are byte-identical; all 12 negative controls rejected their invalid inputs.

**Theorem 5.1.** *Every finite real set $A$ satisfying* $\mathrm{FS}(A) \subseteq 4A$ *has* $|A| \leq 88$. *Consequently the independently closed range is*

$$\boxed{71 \leq N \leq 88}.$$

*Proof.* The preceding independently audited chain gives $|A| \leq 89$. A hypothetical 89-element set can be reduced as in Section 1 and realizes one of the completely reconstructed necessary models. The geometric lemmas above give the new rows simultaneously with the inherited rows. Every parameter and both children of every used disjunction are covered by exact nonnegative integer row combinations giving a strict contradiction. Thus size 89 is impossible. The established 71-element construction supplies the lower bound. □

Scalar, lattice and hole calculations come from the endpoint-specific progress proof, not from an archived endpoint conclusion. The new geometric implications are proved above. The optimizer only proposes multipliers; failed rational reconstructions are rejected. The final verifier uses the standard library and exact integer/rational arithmetic. Actual-set geometry supplements the proofs: 240 finite sets, 2,880 emitted leading-core rows, 1,680 whole-side rows and 240 allocation checks pass. A negative control rejects the double-outside-loss shortcut.

**Provenance and scope.** The leading-core and whole-side ideas occur in the archived size-89 modules `translated_core_overlap89.py` and `whole_side89.py`, in the upper-86 archive [4]. Here their necessary inequalities are rederived and implemented in the current independent model. No archived size-89 certificate, matrix or acceptance status is an input to the new verifier. This work does not claim to improve the archived number 84 or determine the exact maximum.

## References


[1] Sum-of-Subset, *90 to 89 continuation*, `reports/upper89_from90_20261018.tex`.

[2] Sum-of-Subset, *Progress toward cardinality 89*, `reports/upper88_progress_20261018.tex`.

[3] Sum-of-Subset, *Systematic proof audit*, `reports/proof_audit_20261018.md`.

[4] Sum-of-Subset, *Complete upper-86 verification archive*, 5 October 2026, `proofs/exact88/` in the frozen payload.